Use GEK-0710

IC4484 FIELD-WEAKENING SYSTEM

Field weakening is supplied for truck applications where more speed is required than is available with the motor directly across the battery (with 1A closed).

After 1A is closed, card 2 senses the truck speed by sensing current in the series field. After a suitable speed has been attained on full field, field current drops to the set value. FW contactor is then energized to insert field weakening resistor 3RES in parallel with the drive motor field. If current in the series field increases, as when going up a grade, card 2 drops out FW contactor, reapplying full field to the motor.

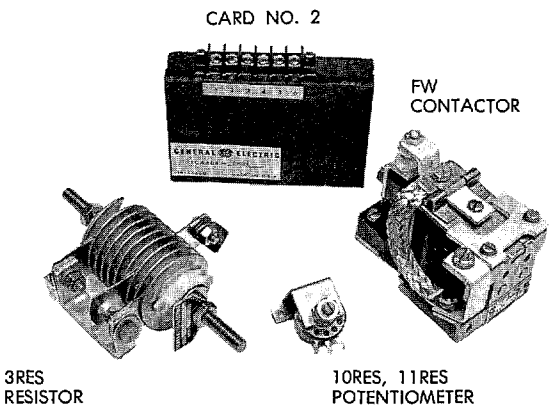


Fig. 1. Field-weakening components

OPERATION

Single Potentiometer Adjustment (Fig. 2)

In this circuit, decreasing the resistance of 10RES lowers the current pickup point of FW (higher truck speed before FW picks up). The current at dropout will be reduced by the same ratio. If additional range of adjustment is required, reconnect 10 RES from terminal 5 to terminal 6 of card 2.

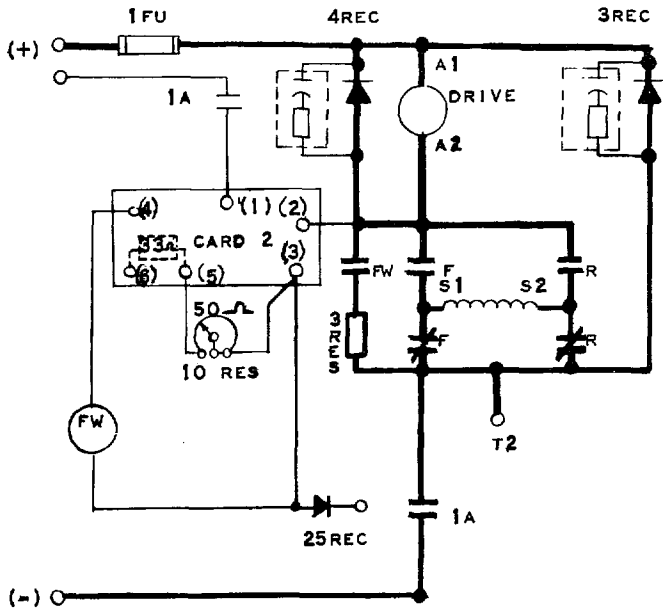


Fig. 2. Single potentiometer adjustment

Double Potentiometer Adjustment (Fig. 3)

In this circuit, 10RES functions as the pickup adjustment; 11RES is the dropout adjustment. Reducing the resistance of these potentiometers will reduce the current at which FW picks up or drops out.

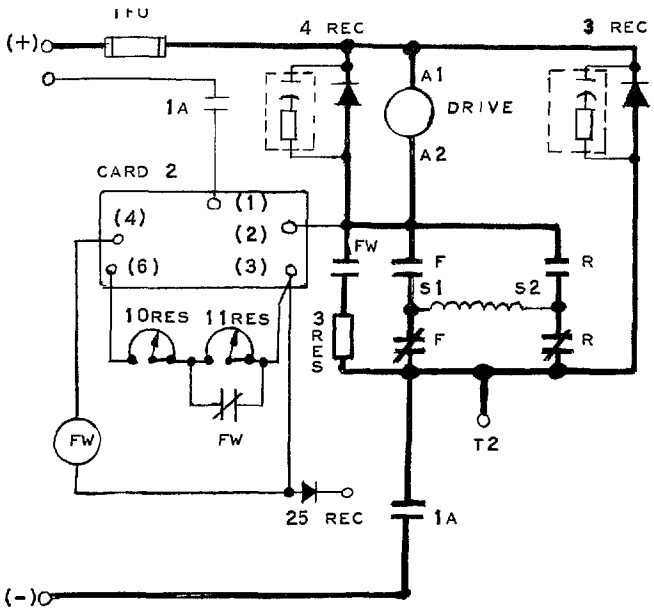


Fig. 3. Double potentiometer adjustment

The information contained herein is intended to assist truck users and dealers in the servicing of control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment or provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

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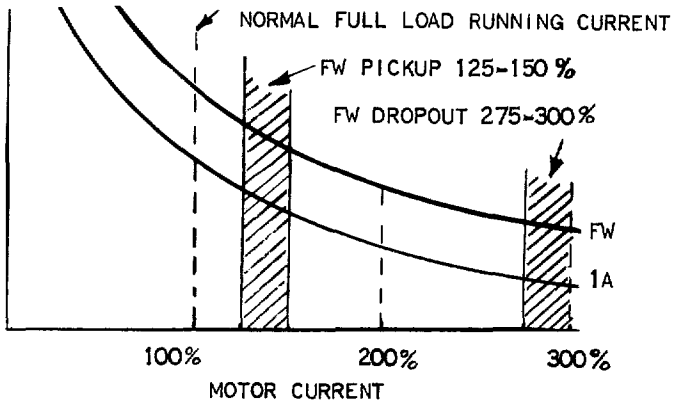


Fig. 4. Pickup and dropout ranges for FW contactor based on full-load level-running speed (1A).

TESTING CARD 2

Disconnect leads to card 2 terminals 1 and 2. Insert paper between power tips of FW contactor. Adjust 10RES to give at least 10 ohms resistance between terminal 5 and terminal 3 (Fig. 2) or terminal 6 and terminal 3 (Fig. 3). Apply positive control volts to terminal 1, FW contactor should not pick up. Short terminals 2 and 3; FW contactor should pick up. If the card does not operate as described, replace the card.

CHECKING OPERATION

To check field weakening connect ammeter and shunt between battery negative and 1REC. Connect battery and depress the accelerator slightly to assure that a positive deflection of the meter is obtained.

Accelerate a loaded truck on the level. FW contactor should close about 1 second after 1A closes when the current reduces to about 125 to 150 - percent of the loaded - level motor current.

Decelerate a loaded truck by going up a ramp or by applying the brake (with the brake switch jumpered out). FW contactor should drop out when current increases to about 300 - percent of loaded - level motor current.

If field weakening is not properly set, the current pickup and dropout range can be adjusted by turning potentiometers 10RES and 11RES as shown in Figs. 2 and 3.

TROUBLE SHOOTING

Symptoms	What To Do
1. Failure of FW contactor to operate.	● Check for control volts from Terminal 1 to Terminal 3 of Card 2 when 1A is closed. If not, check 1A interlock and wiring. Check circuit from Terminal 3 to negative. In Fig. 2 check 1A switchette in accelerator. ● Jumper Terminals 1 and 4 of Card 2. If FW contactor does not pick up, check contactor coil and wiring. ● Check wiring from Terminal 2 of Card 2 to A2. ● Check 10RES and 11RES for open circuit. ● Check card 2.
2. FW contactor operates, but no increase in truck speed is observed.	● Check 3RES for open circuit. ● Check FW power tips.
3. FW contactor operates, but truck speed varies.	● Check for truck grounds around 3RES, or loose connections in FW circuit.

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